SE Lab-7

1. Buggy application output:

```
PS C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7> python process
or.py
==== Running calculate_dimensional_weight ====
Enter dimensions LxWxH in cm (e.g. 20x30x10): 40x75x3
Dimensional Weight (kg): 1.8

==== Running format_tracking_label ====
Enter raw code (e.g. E123): E435
Formatted Label: E-000435

==== Running parse_package_manifest ====
Enter manifest JSON (e.g. {"weight": 12.5}): {"weight": 3}
Manifest Weight: 3.0
```

2. Failing fuzz tests:

```
PS C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7> python -m pytest test_processor.py
========================== test session starts ==========================
platform win32 -- Python 3.12.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15, platformdirs-4.13.0
collected 3 items

test_processor.py FFF                                                  [100%]

================================ FAILURES ================================
_____________ test_calculate_dimensional_weight_no_crash _____________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 5, in test_calculate_dimensional_weight_no_crash
  |     def test_calculate_dimensional_weight_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 7, in test_calculate_dimensional_weight_no_crash
    |     calculate_dimensional_weight(s)
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 6, in calculate_dimensional_weight
    |     width = float(dims[1])
    |                   ~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 9, in test_calculate_dimensional_weight_no_crash
    |     assert False, f"calculate_dimensional_weight crashed with: {e}"
    | AssertionError: calculate_dimensional_weight crashed with: list index out of range
    | assert False
    | Failing test case: test_calculate_dimensional_weight_no_crash(
    |     s='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 7, in test_calculate_dimensional_weight_no_crash
    |     calculate_dimensional_weight(s)
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 5, in calculate_dimensional_weight
    |     length = float(dims[0])
    |              ^^^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 9, in test_calculate_dimensional_weight_no_crash
    |     assert False, f"calculate_dimensional_weight crashed with: {e}"
    | AssertionError: calculate_dimensional_weight crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_calculate_dimensional_weight_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 7, in test_calculate_dimensional_weight_no_crash
    |     calculate_dimensional_weight(s)
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 4, in calculate_dimensional_weight
    |     dims = dimensions_str.split('x')
    |            ^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 9, in test_calculate_dimensional_weight_no_crash
    |     assert False, f"calculate_dimensional_weight crashed with: {e}"
    | AssertionError: calculate_dimensional_weight crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_calculate_dimensional_weight_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
```

```
_____ test_format_tracking_label_safe _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 12, in test_format_tracking_label_safe
  |     def test_format_tracking_label_safe(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 14, in test_format_tracking_label_safe
    |     format_tracking_label(s)
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 12, in format_tracking_label
    |     tracking_num = int(raw_code[1:])
    |                    ^^^^^^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 16, in test_format_tracking_label_safe
    |     assert False, f"format_tracking_label crashed with: {e}"
    | AssertionError: format_tracking_label crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_format_tracking_label_safe(
    |     s='0',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py:15
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 14, in test_format_tracking_label_safe
    |     format_tracking_label(s)
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 11, in format_tracking_label
    |     prefix = raw_code[0].upper()
    |              ~~~~~~~~~~^^^
    | IndexError: string index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 16, in test_format_tracking_label_safe
    |     assert False, f"format_tracking_label crashed with: {e}"
    | AssertionError: format_tracking_label crashed with: string index out of range
    | assert False
    | Failing test case: test_format_tracking_label_safe(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py:15
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 14, in test_format_tracking_label_safe
    |     format_tracking_label(s)
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 11, in format_tracking_label
    |     prefix = raw_code[0].upper()
    |              ~~~~~~~~~~^^^
    | TypeError: 'NoneType' object is not subscriptable
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 16, in test_format_tracking_label_safe
    |     assert False, f"format_tracking_label crashed with: {e}"
    | AssertionError: format_tracking_label crashed with: 'NoneType' object is not subscriptable
    | assert False
    | Failing test case: test_format_tracking_label_safe(
    |     s=None,  # or any other generated value
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py:15
    +------------------------------------
```

```
_____ test_parse_package_manifest_safe _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 19, in test_parse_package_manifest_safe
  |     def test_parse_package_manifest_safe(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 5 distinct failures. (5 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 21, in test_parse_package_manifest_safe
    |     parse_package_manifest(s)
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 16, in parse_package_manifest
    |     data = json.loads(json_payload)
    |            ^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\DELL\AppData\Local\Programs\Python\Python312\Lib\json\__init__.py", line 346, in loads
    |     return _default_decoder.decode(s)
    |            ^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\DELL\AppData\Local\Programs\Python\Python312\Lib\json\decoder.py", line 340, in decode
    |     raise JSONDecodeError("Extra data", s, end)
    | json.decoder.JSONDecodeError: Extra data: line 1 column 2 (char 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 23, in test_parse_package_manifest_safe
    |     assert False, f"parse_package_manifest crashed with: {e}"
    | AssertionError: parse_package_manifest crashed with: Extra data: line 1 column 2 (char 1)
    | assert False
    | Failing test case: test_parse_package_manifest_safe(
    |     s='00',
    | )
```

```
+---------------- 2 ----------------
| Traceback (most recent call last):
|   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 21, in test_parse_package_manifest_safe
|     parse_package_manifest(s)
|   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 16, in parse_package_manifest
|     data = json.loads(json_payload)
|            ^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\DELL\AppData\Local\Programs\Python\Python312\Lib\json\__init__.py", line 346, in loads
|     return _default_decoder.decode(s)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\DELL\AppData\Local\Programs\Python\Python312\Lib\json\decoder.py", line 337, in decode
|     obj, end = self.raw_decode(s, idx=_w(s, 0).end())
|                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\DELL\AppData\Local\Programs\Python\Python312\Lib\json\decoder.py", line 353, in raw_decode
|     obj, end = self.scan_once(s, idx)
|                ^^^^^^^^^^^^^^^^^^^^^^
| json.decoder.JSONDecodeError: Expecting property name enclosed in double quotes: line 1 column 2 (char 1)
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 23, in test_parse_package_manifest_safe
|     assert False, f"parse_package_manifest crashed with: {e}"
| AssertionError: parse_package_manifest crashed with: Expecting property name enclosed in double quotes: line 1 column 2 (char 1)
| assert False
| Failing test case: test_parse_package_manifest_safe(
|     s='{',
| )
+---------------- 3 ----------------
| Traceback (most recent call last):
|   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 21, in test_parse_package_manifest_safe
|     parse_package_manifest(s)
|   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 17, in parse_package_manifest
|     return float(data["weight"])
|                  ~~~~^^^^^^^^^^
| TypeError: 'int' object is not subscriptable
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 23, in test_parse_package_manifest_safe
|     assert False, f"parse_package_manifest crashed with: {e}"
| AssertionError: parse_package_manifest crashed with: 'int' object is not subscriptable
| assert False
| Failing test case: test_parse_package_manifest_safe(
|     s='0',
| )
+---------------- 4 ----------------
| Traceback (most recent call last):
|   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 21, in test_parse_package_manifest_safe
|     parse_package_manifest(s)
|   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 16, in parse_package_manifest
|     data = json.loads(json_payload)
|            ^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\DELL\AppData\Local\Programs\Python\Python312\Lib\json\__init__.py", line 346, in loads
|     return _default_decoder.decode(s)
|            ^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\DELL\AppData\Local\Programs\Python\Python312\Lib\json\decoder.py", line 337, in decode
|     obj, end = self.raw_decode(s, idx=_w(s, 0).end())
|                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\DELL\AppData\Local\Programs\Python\Python312\Lib\json\decoder.py", line 355, in raw_decode
|     raise JSONDecodeError("Expecting value", s, err.value) from None
| json.decoder.JSONDecodeError: Expecting value: line 1 column 1 (char 0)
|
| During handling of the above exception, another exception occurred:
|
| Traceback (most recent call last):
|   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 23, in test_parse_package_manifest_safe
|     assert False, f"parse_package_manifest crashed with: {e}"
| AssertionError: parse_package_manifest crashed with: Expecting value: line 1 column 1 (char 0)
| assert False
| Failing test case: test_parse_package_manifest_safe(
|     s='',  # or any other generated value
| )
```

```
    +---------------- 5 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 21, in test_parse_package_manifest_safe
    |     parse_package_manifest(s)
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\processor.py", line 16, in parse_package_manifest
    |     data = json.loads(json_payload)
    |            ^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\DELL\AppData\Local\Programs\Python\Python312\Lib\json\__init__.py", line 339, in loads
    |     raise TypeError(f'the JSON object must be str, bytes or bytearray, '
    | TypeError: the JSON object must be str, bytes or bytearray, not NoneType
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7\test_processor.py", line 23, in test_parse_package_manifest_safe
    |     assert False, f"parse_package_manifest crashed with: {e}"
    | AssertionError: parse_package_manifest crashed with: the JSON object must be str, bytes or bytearray, not NoneType
    | assert False
    | Failing test case: test_parse_package_manifest_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
========================= short test summary info ==========================
FAILED test_processor.py::test_calculate_dimensional_weight_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_format_tracking_label_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_processor.py::test_parse_package_manifest_safe - ExceptionGroup: Hypothesis found 5 distinct failures. (5 sub-exceptions)
============================ 3 failed in 1.52s =============================
```

3. Fixed application output:

```
PS C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7> python processor.py
==== Running calculate_dimensional_weight ====
Enter dimensions LxWxH in cm (e.g. 20x30x10): 40x75x3
Dimensional Weight (kg): 1.8

==== Running format_tracking_label ====
Enter raw code (e.g. E123): E435
Formatted Label: E-000435

==== Running parse_package_manifest ====
Enter manifest JSON (e.g. {"weight": 12.5}): {"weight": 3}
Manifest Weight: 3.0
```

4. Passing fuzz tests:

```
PS C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7> python -m pytest test_processor.py
============================= test session starts =============================
platform win32 -- Python 3.12.7, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\DELL\Documents\L6_L7_04_shipping_quotation\Lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15, platformdirs-4.13.0
collected 3 items

test_processor.py ...                                                    [100%]

============================== 3 passed in 0.79s ==============================
```